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Homework 11
MATH 307

1. a)

$$\det(A) = \det\left(\begin{bmatrix} a & 1 & 0 \\ 0 & 1 & b \\ 0 & 0 & c \end{bmatrix}\right) = c \cdot \det\left(\begin{bmatrix} a & 1 \\ 0 & 1 \end{bmatrix}\right) - 0 + 0 = c \cdot (a(1) - 0) = \boxed{ca}$$

b) When $\boxed{a = 1,\ b = 0,\ \text{and}\ c = 1}$ or any $a, c \neq 0$ A is invertible because $\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$ are linearly independent

$$\det(A^{-1}) = \frac{1}{\det(A)} = \boxed{\frac{1}{ac}}$$

2. I will find $\det(-2A)$ in terms of $\det(A)$

$$-2A = [-2a_1, -2a_2, \dots -2a_7] \Leftrightarrow \det(-2A) = \det([-2a_1, -2a_2, \dots -2a_7]) = -2\det([a_1, -2a_2, \dots -2a_7]) = (-2)^7 \det([a_1, a_2, \dots, a_7])$$

$$\boxed{\det(-2A) = -128 \det(A)}$$

3. Q is invertible, $Q^T = Q^{-1}$

$\det(Q) = \det(Q^T)$, $\det(Q) = \frac{1}{\det(Q^{-1})} = \frac{1}{\det(Q^T)} = \frac{1}{\det(Q)}$

$\underline{\det(Q) = \frac{1}{\det(Q)}}$, $\boxed{\det(Q) = 1 \text{ or } -1}$

4. U, V are orthogonal, $\det(U) = \det(V^T) = 1$ or -1 as shown in 3.

$$|\det(A)| = |\det(U \Sigma V^T)| = |\det(U)\det(\Sigma)\det(V^T)| = |\det(\Sigma)|$$

$$|\det(A)| = |\det(\Sigma)| = \left|\det\left(\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}\right)\right| = |\sigma_1 \det\left(\begin{bmatrix} 1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}\right)| = |\sigma_1 \sigma_2 \cdots \sigma_n \det(I_n)| = |\sigma_1 \sigma_2 \dots \sigma_2| =$$

all positive by definition of SVD

$$\boxed{|\det(A)| = \sigma_1 \sigma_2 \cdots \sigma_n}$$

5. $\det(A) = \det(SBS^{-1}) = \det(S)\det(B)\det(S^{-1})$

$\det(S^{-1}) = \frac{1}{\det(S)}$

$\det(A) = \det(S)\det(B) \cdot \frac{1}{\det(S)}$

$$\boxed{\det(A) = \det(B)}$$

6. Every SPD has a Cholesky Factorization: $A = RR^T$

$\det(A) = \det(RR^T) = \det(R)\det(R^T)$

$\det(R) = \det(R^T)$

$\det(A) = (\det(R))^2 > 0$ ← squared = positive

$\boxed{\det(A) > 0}$

7. a) The determinant of Q is either $\boxed{1 \text{ or } -1}$ as shown in 3.

b) $\det(A) = \det(QR) = \det(Q)\det(R) = \underline{\det(R) \text{ or } -\det(R)}$

i.e. $\boxed{|\det(A)| = |\det(R)|}$

c) $\det(A) = \det(A^T)$

$\det(AA^T) = \det(A)\det(A^T) = (\det(A))^2$

$|\det(A)| = |\det(R)|$, $(\det(A))^2 = (\det(R))^2$

$\boxed{\det(AA^T) = (\det(R))^2}$

d) If $\det(R) = 3$, $\underline{\det(A) = 3 \text{ or } -3}$ because $\det(A) = \det(Q)\det(R)$

If $\det(A) \neq 0$, then $\underline{A \text{ is invertible}}$ because it has linearly independent columns

$\det(A) = \frac{1}{\det(A^{-1})}$, $|3| = \frac{1}{|\det(A^{-1})|}$

$\boxed{|\det(A^{-1})| = \frac{1}{3}}$